Lab 20 — Backup and Recovery with restic

You will configure encrypted, deduplicated backups, simulate ransomware, and perform a point-in-time restore.

Run on https://killercoda.com/playgrounds/scenario/ubuntu

Required software (free, apt): - `restic`

Step 1 — Install

```
apt update && apt install -y restic
```

```
restic version
```

Step 2 — Create a "production" data set and a backup repository

```
mkdir -p /data /backup
```

```
for i in $(seq 1 5); do echo "important record $i" > /data/rec$i.txt; done
```

```
export RESTIC_PASSWORD='Lab1234!'
```

```
restic -r /backup init
```

The repo is encrypted with the password — even if the backup disk is stolen the data is safe.

Step 3 — Take the first snapshot

```
restic -r /backup backup /data
```

```
restic -r /backup snapshots
```

Note the snapshot ID — that is your **RPO checkpoint**.

Step 4 — Modify data, take a second snapshot

```
echo "edit" >> /data/rec1.txt
```

```
echo "new record" > /data/rec6.txt
```

```
restic -r /backup backup /data
```

```
restic -r /backup snapshots
```

Step 5 — Simulate ransomware

```
for f in /data/*.txt; do echo "ENCRYPTED_BY_RANSOMWARE" > "$f"; done
```

```
ls -l /data && cat /data/rec1.txt
```

Step 6 — Restore from the previous snapshot

```
SNAP=$(restic -r /backup snapshots --json | python3 -c "import
sys,json; print(json.load(sys.stdin)[-2]['id'])")
```

```
echo "Restoring snapshot $SNAP"
```

```
restic -r /backup restore $SNAP --target /restore
```

```
diff -r /restore/data /data
```

```
ls /restore/data
```

```
cat /restore/data/rec1.txt
```

You have just demonstrated **RTO** (time to recover) and **RPO** (data loss window).

Step 7 — Apply 3-2-1 strategy

	Where	What
1	`/data` (production)	Live data
2	`/backup` (local)	Same machine snapshot
3	Off-site (S3, B2)	`restic -r s3:...`

Initialise an off-site repo (example with the free Backblaze B2 tier):

```
# Replace with your B2 credentials, or use a local minio for testing
```

```
# export B2_ACCOUNT_ID=... B2_ACCOUNT_KEY=...
```

```
# restic -r b2:bucket:/path init
```

Free online tools

- **Backblaze B2 (10 GB free)** — https://www.backblaze.com/b2
- **Wasabi** (S3-compatible cold storage) — https://wasabi.com
- **MinIO (self-hosted S3)** — https://min.io
- **rclone** (sync/encrypt) — https://rclone.org
- **NIST SP 800-34 Contingency Planning** — https://csrc.nist.gov/publications/detail/sp/800-34/rev-1/final

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What you learned

- restic combines encryption + deduplication + snapshots in one binary.
- Why the 3-2-1 rule survives ransomware, hardware failure, and regional outage.
- How to measure RTO and RPO from real restore timing.